Team Netherite

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ANLP Final

Evaluating Multilingual Reasoning through a Chain-of-Thought Lens

Final presentation

English vs Hindi reasoning on date and time questions

Questions, Sarvam-105B

English
Today is 2025-08-26, what is
the date going to be in 3 days?

Hindi
आज 2025-08-26 है, 3 दिनों में
तिथि क्या होगी?

Varied

Prompt
direct or chain-of-thought (CoT)

Reasoning language
English, Hindi or Hinglish

Distractor
none, time-related or unrelated

Measured

Accuracy
vs gold answers

Reasoning traces
language, steps, checks

Errors
error types by condition

Does CoT help temporal reasoning equally in English and Hindi?

We found no study covering all five

	CoT	Distractors	Language of reasoning	Dates and times	Hindi
Wei et al.; Sprague et al. CoT helps most on math and symbolic tasks	●	○	○	○	○
Shi et al. (GSM-IC); Mirzadeh et al. Irrelevant text hurts English arithmetic	◐	●	○	○	○
Shi et al. (MGSM) Native vs English CoT, ten other languages	●	○	●	○	○
Mazzia et al. (TRD) Dates and times, ten languages, CoT turned off	○	●	○	●	●
This work All five on the same questions	●	●	●	●	●

● covered ◐ used, but not compared on and off under the same distractor

Research questions

RQ1 Does CoT help, and does that depend on language and difficulty?

RQ2 Does CoT make distractors more harmful?

RQ3 How much of the English-Hindi gap follows the language of reasoning?

RQ4 Does asking for Hinglish reasoning help?

RQ5 Does divergence between English and Hindi traces track disagreement?

RQ6 Can we say that distractors in CoT traces cause errors?

6,750 questions in each language

9	×	5	×	150	=	6,750	13,500
tasks		difficulty levels		questions per cell		questions per language	gold answers rechecked with datetime

Grid

Levels	short, medium, long, very long, very very long
Languages	en_US, hi_IN, same question IDs
Input variants	no distractor, time-related, unrelated
Prompts per condition	6,750 × 3 variants × 2 languages = 40,500
Direct + CoT	81,000 responses
Generator	custom-generator.py, seed 9, up to 10 draws per sample, duplicates discarded

Nine tasks with a fixed answer form

Task	Question (level: long)	Gold	Format
Date addition	Today is 2030-03-12, what is the date going to be in 13 days?	2030-03-25	YYYY-MM-DD
Date subtraction	Today is 2027-02-02, what was the date 9 days ago?	2027-01-24	YYYY-MM-DD
Date duration	How many day(s) (e.g., 5, 9) have passed between 2028-05-28 and 2028-06-12?	15	whole days
Date recurrence	Today is 2027-02-19, and I have a recurrence every 11 days. Without counting today, what is the date of the 5 occurrence?	2027-04-15	YYYY-MM-DD
Interval date	If today is 2026-12-22, what are the dates for the start and end of next week, assuming Monday is the first day of the week?	2026-12-28 to 2027-01-03	A to B
Day of week	What day of the week (e.g., Monday, Tuesday, ...) is 2025-07-31?	Thursday	weekday name
Time addition	It is now 07:09, what will the time be in 4 hours and 17 minutes?	11:26	HH:MM
Time subtraction	It is now 15:53, what was the time 0 hours and 53 minutes ago?	15:00	HH:MM
Time duration	If I looked at the clock at 01:45 and now it is 05:17, how many minutes (e.g., 15, 23) have gone by?	212	whole minutes

Each question also runs with two distractors

Time-related: 35 sentence pairs

I always get confused with months that have 31 days.
मुझे 31 दिनों वाले महीनों में भ्रम होता है।

August is my favorite month of the year.
अगस्त मेरा पसंदीदा महीना है।

I always forget how many days are in each month.
मैं भूल जाता हूं कि हर महीने में कितने दिन होते हैं।

It's interesting how we measure time in different cultures.
यह दिलचस्प है कि हम समय कैसे मापते हैं।

The summer of 2025 has been quite interesting.
2025 की गर्मी दिलचस्प रही।

Time really flies when you're having fun.
मज़े करते समय वक्त उड़ जाता है।

Unrelated: 43 sentence pairs

The garden needs new flowers.
बगीचे में नए फूल चाहिए।

My brother bought a motorcycle.
मेरे भाई ने मोटरसाइकिल खरीदी।

The recipe calls for sugar.
रेसिपी में चीनी चाहिए।

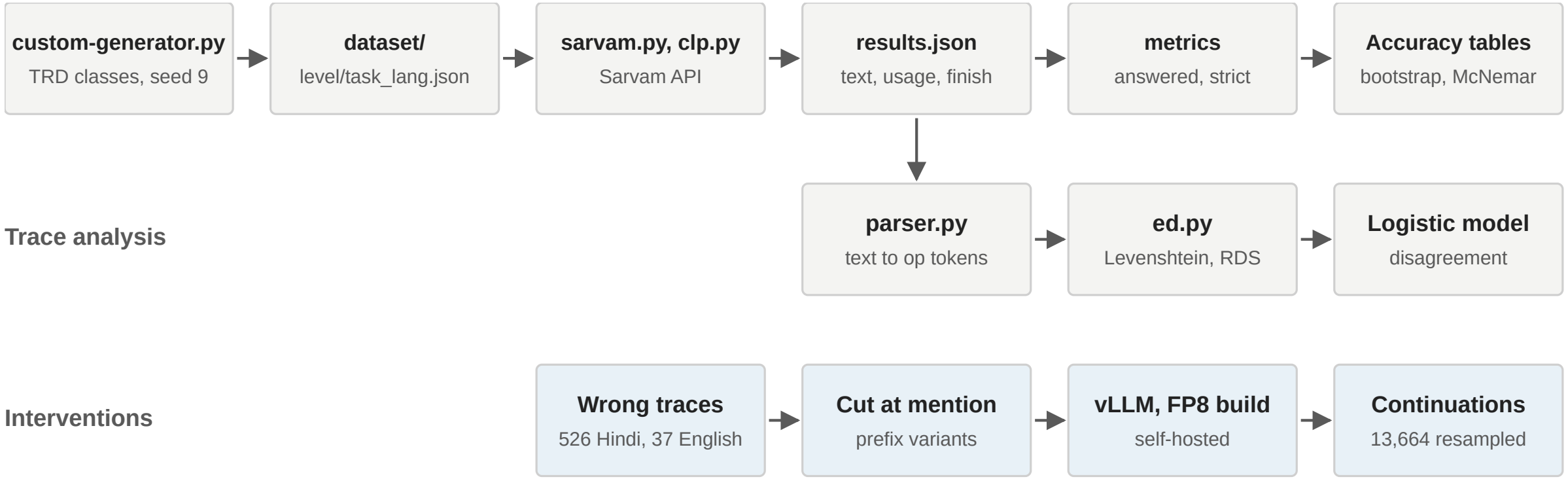
Classical music helps me concentrate.
शास्त्रीय संगीत मदद करता है।

The museum opens at nine.
संग्रहालय नौ बजे खुलता है।

I need new running shoes.
मुझे नए दौड़ने के जूते चाहिए।

Per task: 5 sentences of each kind, 45 task-sentence units per kind, 90 in all. Prepended to the question.

Code pipeline



Prompting conditions

	Reasoning language: No written reasoning	English	Hindi	Hinglish mix
English question	**Direct** 6,750 questions	**CoT** 6,750 questions	**Not run**	**Hinglish** 2,250 questions
Hindi question	**Direct** 6,750 questions	**en-CoT, CLP** 2,250 questions	**CoT** 6,750 questions	**Hinglish** 2,250 questions

Prompts

Template, all conditions

{mode} Write your entire response in {language}. End your response with a final line of the form "Answer: <answer>", where <answer> is {format}. Keep the word "Answer:" in English, use Western digits (0-9), and put nothing else on that line.

Direct

Answer the following question directly. Do not provide any reasoning or explanation. Give only the final answer.

CoT

Solve the following question by reasoning through it step by step. Your reasoning must be written entirely in {language}; do not switch to another language at any point.

en-CoT

The following question is written in {source}. Solve it by reasoning through it step by step in {language}. Your reasoning must be written entirely in {language}; do not switch to another language at any point.

Hinglish

Solve the following question by reasoning through it step by step. Your reasoning must be written in Hinglish, naturally mixing Hindi and English using only the Roman (Latin) script. Do not write in Devanagari script (the final "Answer:" line is the only exception).

CLP prompts

Turn 1: understand

Please act as an expert in multi-lingual understanding in {source}. Write your entire response in {target}. Request: {question} Let's understand the task in {target} step-by-step!



Turn 2: solve, same conversation

After understanding, you should act as an expert in {task} in {target}. Let's resolve the task you understand above step-by-step! Your reasoning must be written entirely in {target}; do not switch to another language at any point. End your response with a final line of the form "Answer: <answer>", where <answer> is {format}.

Source, target	hi_IN, en_US
Task	temporal reasoning
Changes from Qin et al.	temperature fixed at 0; added write-in-English rule; no CLSP voting

Response sets and settings

Set	Prompt	Q. language	Questions	Responses	Thinking	Max tokens
Standard	Direct, CoT	both	6,750	81,000	off	4,096
Thinking-on	Direct, CoT	both	450	5,400	low	16,000
en-CoT	English CoT	Hindi	2,250	6,750	off	4,096
CLP	two turns	Hindi	2,250	6,750	off	4,096 each
Hinglish	mixed CoT	both	2,250	13,500	off	4,096
Interventions	continuations	both		13,664	off	

Setting	
Model	sarvam-105b
Temperature, top_p	0.0, 1.0 (resampling 0.8)
Retries	6, on 429, 500, 502, 503, 504
Concurrency, timeout	6, 180 s

One question, every condition

English Today is 2027-05-16, and I have a recurrence every 13 days. Without counting today, what is the date of the 7 occurrence?

Hindi आज 2027-05-16 है, और मेरे पास हर 13 दिनों पर एक पुनरावृत्ति है। आज को छोड़कर, 7वीं आवृत्ति की तिथि क्या है?

Distractor The last time I had a weekly commitment was in January. / आखिरी साप्ताहिक प्रतिबद्धता जनवरी में थी।

Gold 2027-08-15 Long, date recurrence, ID 12. 2027-08-02 adds six intervals, not seven.

Condition	Reasons in	No distractor	Time-related
English direct		2027-06-05 ×	2027-04-19 ×
English CoT	English	2027-08-15 ✓	cut off
Hindi direct		2027-07-27 ×	2027-03-06 ×
Hindi CoT	Hindi	2027-08-02 ×	2027-04-20 ×
en-CoT (Hindi Q.)	English	2027-08-15 ✓	cut off
CLP (Hindi Q.)	English	2027-08-02 ×	2027-08-02 ×
Hinglish (English Q.)	Roman mix	2027-08-02 ×	2027-08-01 ×
Hinglish (Hindi Q.)	Devanagari mix	2027-08-02 ×	2027-07-06 ×

Parser: trace text to op tokens



Value	In text	Token
Date	2025-3-7, Mar 7, 2025, 7 मार्च	2025-03-07, or 03-07 if partial
Time	9:05, 1:23 PM	09:05, 13:23
Weekday	Thu, गुरुवार, वीरवार	Thursday
Month	March 2031, मार्च	2031-03, 03
Number	42, twenty-one, दो हज़ार पच्चीस	42, 21, 2025

Coverage, precision	
Standard CoT	95-96%
Thinking reasoning	86-88%
Op right	85%
Op + arguments	77%

Precision: 240 tokens, one LLM annotator

Parser ops

Op	Args	Meaning
GIVEN	value	Read a value from question
GOAL	target	Restate what must be found
PLAN	method	Choose a solution method
RULE	rule	Apply a stated task rule
LEAP	year	Decide whether year is leap
MONTHLEN	month	Look up days in month
ANCHOR	date, weekday	Recall a date's weekday
CONVERT	value, unit	Change units or representation
ADD	x, y	Add two values
SUB	x, y	Subtract one value from another
MUL	x, y	Multiply two values
DIV	x, y	Divide, remainder or modulo

Op	Args	Meaning
CMP	x, y	Compare two values
ROLL	unit	Carry across day, month or year
STEP	value, delta	Move one unit at once
CUMADD	amount	Add to a running total
CUMSUB	amount	Subtract from a running total
CHECK		Verify an earlier result
REVISE		Correct or abandon a step
IGNORE		Mention or dismiss the distractor
FORMAT		Deliberate over format or language
ANSWER	value	State the final answer
SPITANSWER	value	Write the final answer line

RQ6 interventions

Hindi trace, English translation. August is my favourite month. Today is 2026-07-24; what will the date be in 8 days?

s0 First, we have to understand the given information carefully.

s1 Today's date is 2026-07-24.

s2 The question asks what the date will be 8 days from today.

s3 (k) The question also says that August is my favourite month.

Variant		Prefix kept				Question
P1	one sentence earlier	s0	s1	s2	s3 (k)	with distractor
P	just before the mention	s0	s1	s2	s3 (k)	with distractor
M	including the mention	s0	s1	s2	s3 (k)	with distractor
Q	same prefix as P	s0	s1	s2	s3 (k)	**distractor removed**

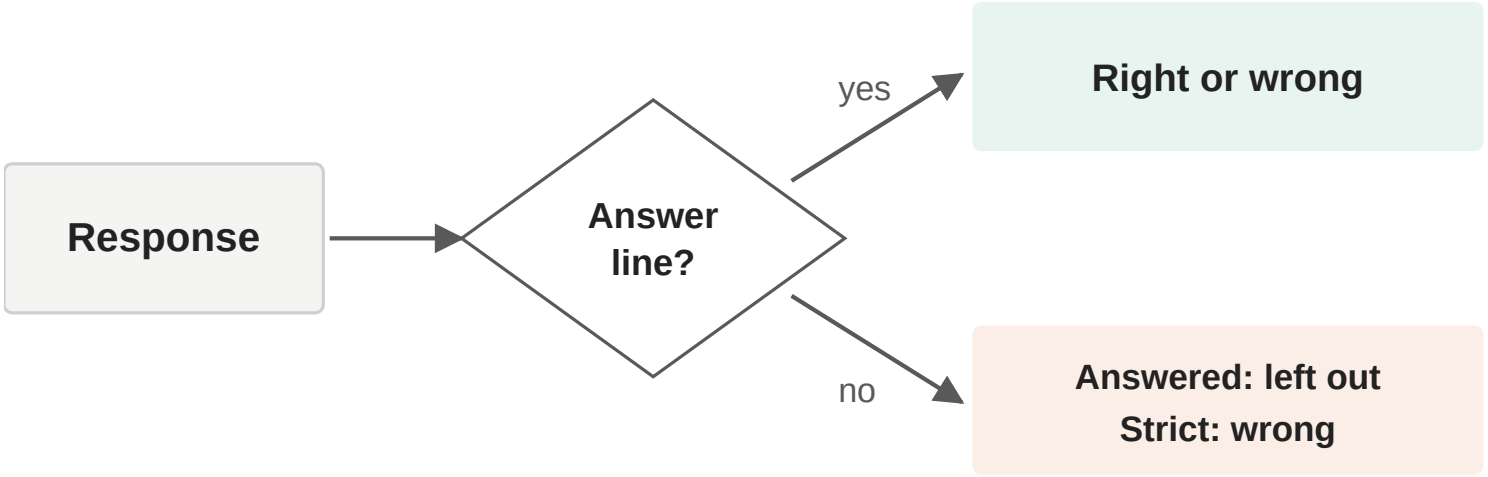
Q − P distractor in the question

M − P written mention

P − P1 ordinary sentence

Originally wrong traces with a time-related mention: 526 Hindi, 37 English. 7 continuations per start, temperature 0.8.

Accuracy: answered and strict



6,750 questions	Answered	Strict
English CoT	88.5	85.7
Hindi CoT	59.4	59.0
English direct	59.0	58.9
Hindi direct	54.7	54.7

answered accuracy = correct / responses with an answer line

strict accuracy = correct / all responses

Gain, gap, drop, penalty, share

Metric	Formula, percentage points	Example
CoT gain	acc(CoT) − acc(direct)	English: 88.5 − 59.0 = +29.5
Gap	acc(English) − acc(Hindi)	CoT: 88.5 − 59.4 = 29.1
Drop	acc(no distractor) − acc(distractor)	Hindi CoT, time-related: 2.6
CoT penalty	drop(CoT) − drop(direct)	Hindi: 2.6 − 0.4 = 2.2
Reasoning part	acc(en-CoT) − acc(Hindi CoT)	82.9 − 59.4 = 23.5
Question part	acc(English CoT) − acc(en-CoT)	88.9 − 82.9 = 6.0
Reasoning share	reasoning part / gap	23.5 / 29.5 = 79.7%

Trace measures

$$\text{TLR} = \text{Devanagari words} / (\text{Devanagari words} + \text{Latin words})$$

Verification regex match on checking phrases

Mention regex match on distractor words; placebo is the same question without it

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Mention odds ratio, per task and level stratum

	Correct	Wrong
Mentions	**a**	**b**
No mention	**c**	**d**

$$\text{OR} = \Sigma\,(a \cdot d / n) \,/\, \Sigma\,(b \cdot c / n)$$

OR < 1: traces with a mention are correct less often.

Reasoning divergence score

a, b: op-token lists of length n, m

$$D(i, 0) = i,\ D(0, j) = j$$

$$D(i, j) = \min \{ D(i-1, j) + 1,\ D(i, j-1) + 1,\ D(i-1, j-1) + \text{cost}(a_i, b_j) \}$$

$$RDS = D(n, m) / \max(n, m)$$

$\text{cost}(a_i, b_j)$	
same token	0
one token has a partial date, the other the full date	0.5
otherwise	1

RDS = 0: same steps, same order

RDS = 1: no step in common

RDS on a real pair

Both correct: "Today is 2025-09-11, what is the date going to be in 2 days?"

English	`GOAL DATE`	`GIVEN 2025-09-11`	`ADD 2025-09-12 1`	`ADD 2025-09-13`	`ANSWER 2025-09-13`	`SPITANSWER 2025-09-13`
Hindi	`gap`	`GIVEN 2025-09-11`	`gap`	`ADD 2025-09-13 2`	`ANSWER 2025-09-13`	`SPITANSWER 2025-09-13`
	edit	match	edit	edit	match	match

RDS = edits / longer length = 3 / 6 = 0.50

0 = same steps in the same order
1 = no steps in common

Answer tokens count same pair, another final answer: 5 / 6 = 0.83

On real pairs new Hindi answers on 11,309 both-correct pairs: +0.082, 68% of the 0.120 gap

RQ5: logistic regression

20,662 English-Hindi trace pairs, 19,471 with thinking off

Hindi only 516, both wrong 1,711



$$\text{logit } P(\text{disagree}) = \beta_0 + \beta_1 \cdot \text{RDS} + \text{task} + \text{difficulty} + \text{distractor} + \text{thinking mode}$$

$$\text{second model adds } \beta_2 \cdot |\text{len en} - \text{len hi}| \,/\, \max(\text{len en}, \text{len hi})$$

$$\text{OR per 0.1 RDS} = \exp(0.1 \cdot \beta_1) = \exp(0.1 \cdot 15.94) = 4.92$$

Standard errors clustered by question. 21,600 pairs loaded, 938 dropped (answer unparsed), 6,739 questions.

RQ6: contrasts

Contrast, accuracy in points	Tests	Hindi
Q − P	distractor in the question	+5.3
M − P	written mention	−3.3
P − P1	ordinary preceding sentence	−0.7
(M − P) − (P − P1)	mention beyond an ordinary sentence	−4.6
B − C	deleting mention vs neutral sentence	+0.0

excess fraction = (wrong(P) − wrong(Q)) / wrong(P) = (60.05 − 54.78) / 60.05 = 8.8%

Statistical tests

Bootstrap intervals

Resample questions; each appears in three distractor conditions.

Q3 | Q1 | Q3

B = 2,000

95% interval = 2.5th and 97.5th percentiles

Exact McNemar

Same question under two conditions. Only disagreeing pairs count.

b: only A right | c: only B right

b ~ Binomial(b + c, 0.5)

Holm correction

Nine tasks tested at once. Sort p-values, smallest first.

$p(i) \times (9 - i + 1)$

Stop at the first p(i) above 0.05.

Answers to the research questions

RQ1	Yes for English, little for Hindi.	**+29.5 vs +4.7 points**
RQ2	Slightly in Hindi, largely through one sentence.	**Penalty 2.2 points**
RQ3	Most of it, descriptively.	**23.5 of 29.5 points**
RQ4	No. Hindi questions don't change, and English questions get worse.	**−0.0 and −18.0 points**
RQ5	Yes as an association, though much of it seems to come from the answers.	**OR 4.92 per 0.1 RDS**
RQ6	No. Removing the distractor helps a little.	**+5.3 points**

Accuracy by set: English CoT is highest in every row

2,250 questions, answered	En direct	En CoT	Hi direct	Hi CoT	CLP	en-CoT	Hingl. En	Hingl. Hi
Date addition	85.6	98.0	82.8	85.1	92.7	95.7	87.3	85.9
Date subtraction	86.0	95.2	84.8	66.0	86.9	94.0	70.7	71.0
Date duration	84.0	99.3	78.9	88.2	92.7	97.7	92.9	82.8
Date recurrence	22.1	90.3	2.8	20.1	26.9	70.3	53.8	18.9
Day of week	14.9	69.5	15.2	21.0	31.8	52.5	34.7	23.5
Interval date	43.2	50.7	46.8	23.9	35.3	44.1	38.5	23.6
Time addition	70.5	95.9	75.6	73.5	88.2	92.6	82.9	72.3
Time subtraction	64.7	97.7	64.9	75.2	90.1	95.7	85.3	74.2
Time duration	65.7	98.9	44.7	80.1	86.7	98.1	90.0	80.9
Short	77.8	93.2	73.9	69.5	77.2	83.0	82.9	66.5
Medium	69.6	91.6	65.3	65.4	75.7	84.9	78.8	67.4
Long	61.2	89.3	57.9	60.4	71.3	84.4	70.4	60.9
Very long	54.3	87.2	48.4	53.7	65.8	83.5	64.0	53.7
Very very long	35.5	82.9	30.3	47.9	61.5	78.5	58.2	48.3
No distractor	60.6	89.3	56.0	60.2	69.3	84.1	70.6	61.0
Time-related	59.9	88.2	54.9	58.9	69.3	81.0	69.7	59.1
Unrelated	58.6	89.2	54.6	59.1	72.3	83.6	72.3	58.0
All, answered	59.7	88.9	55.2	59.4	70.3	82.9	70.9	59.4
All, strict	59.6	86.0	55.2	59.0	69.9	80.8	70.3	59.1

RQ1: Direct answers score 59.0 in English, 54.7 in Hindi

Direct answers

	English	Hindi
Accuracy	59.0	54.7
Date recurrence wrong	80%	97%
Day of week: top weekday	Friday 34.4%	Sunday 58.7%
Wrong dates off by 10, 20 or 30 days	35%	30%
Wrong times off by whole hours	65%	57%
Wrong answers reusing distractor values	6.1%	11.3%
Placebo	3.4%	4.1%
p	0.004	<0.001

Day of week

85% wrong (3,839 of 4,500)

chance: 85.7%

Off by 10

36 days after 2029-03-06

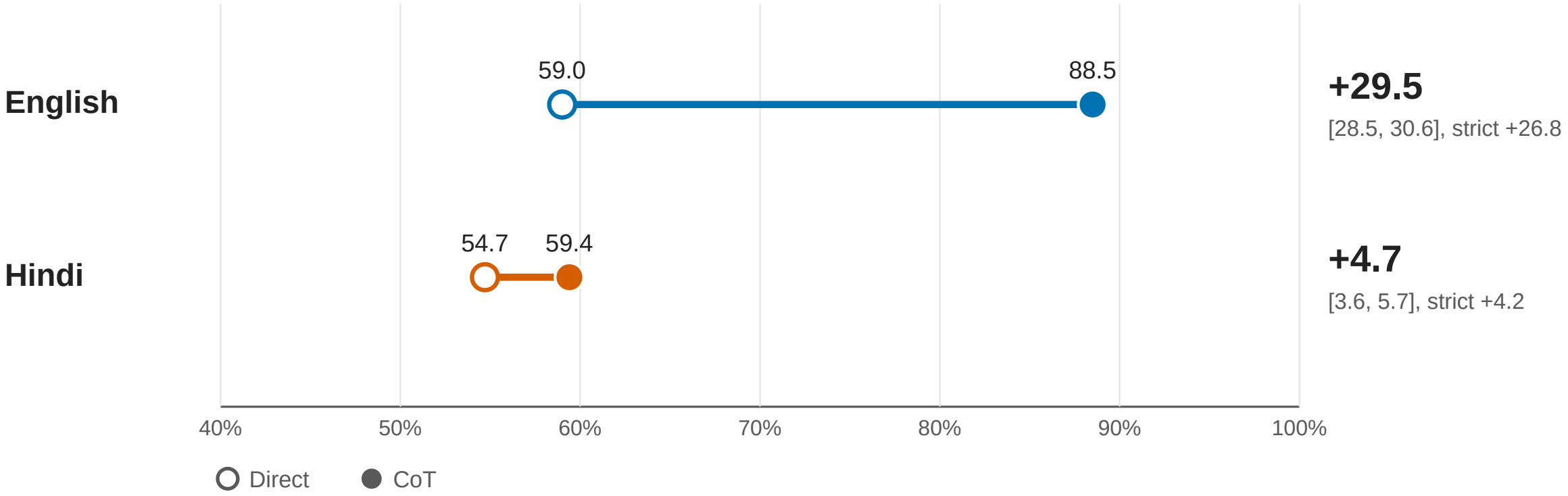
Model 2029-04-01, gold 2029-04-11

Rate under English CoT: 3%

Interval date

22% of wrong English answers give this week

RQ1: CoT helps English far more than Hindi



	None	Time-related	Unrelated
English gain	+28.8	+29.1	+30.7
Hindi gain	+5.6	+3.4	+5.0
Gap, CoT	+28.0	+29.7	+29.7
Gap, direct	+4.8	+4.0	+3.9

Gap direct 4.3, CoT 29.1 (strict 26.7)

Pooled over 3 distractor conditions

RQ1: English gains on every task, Hindi falls on two

	English				Hindi			
CoT gain, 6,750 questions	Pooled	None	Time	Unrel.	Pooled	None	Time	Unrel.
Date addition	+11.8	+11.5	+11.6	+12.4	+3.1	+2.8	+3.8	+2.8
Date subtraction	+7.0	+7.5	+9.2	+4.4	−17.6	−19.9	−15.4	−17.4
Date duration	+16.1	+16.9	+14.7	+16.7	+10.2	+15.4	+6.6	+8.5
Date recurrence	+68.5	+72.3	+63.9	+68.8	+18.0	+20.8	+16.3	+16.8
Day of week	+55.1	+53.8	+55.2	+56.2	+6.4	+7.1	+4.7	+7.5
Interval date	+9.4	+7.2	+4.0	+17.0	−21.8	−23.3	−23.2	−18.9
Time addition	+27.5	+22.9	+29.0	+30.6	+0.5	+0.4	−0.1	+1.1
Time subtraction	+31.1	+29.3	+30.1	+33.9	+8.5	+7.6	+9.5	+8.3
Time duration	+35.3	+32.5	+40.7	+32.8	+33.0	+37.7	+26.2	+35.0

RQ1: Hindi falls on subtraction, interval date and short time tasks

English: CoT minus direct

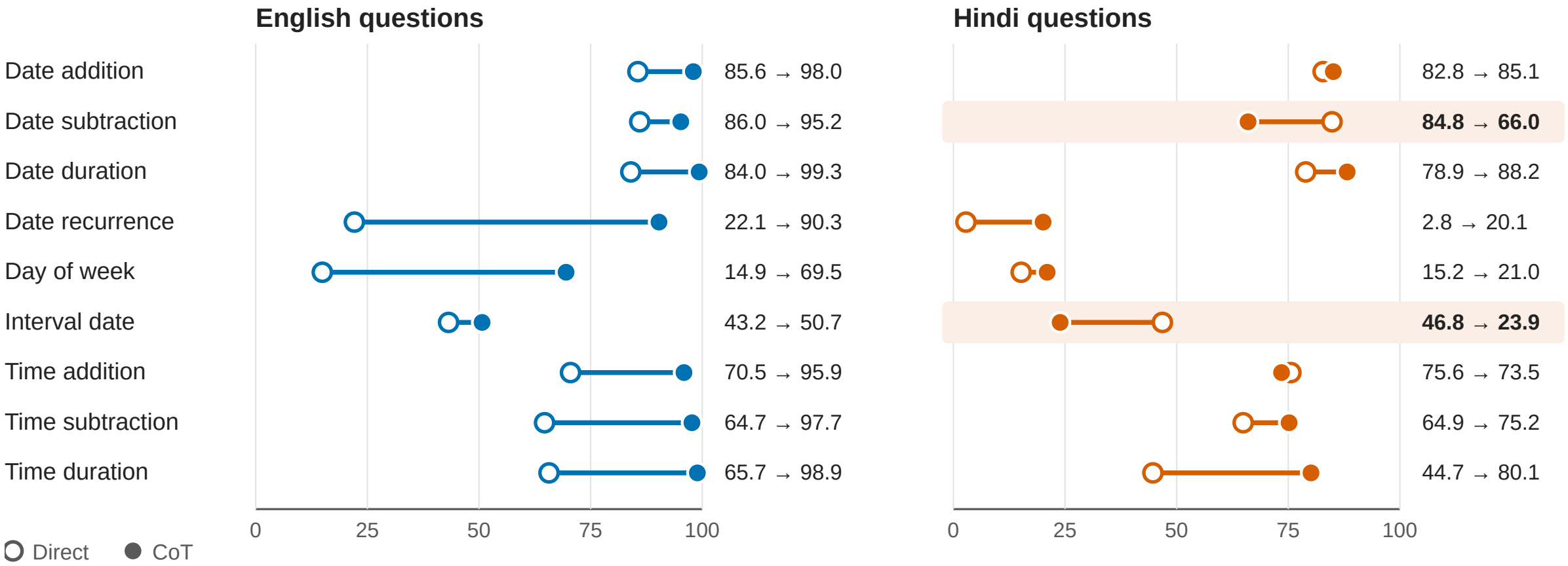
	Short	Med.	Long	V. long	VV. long
Date addition	**+2**	**+3**	**+5**	**+10**	**+40**
Date subtraction	+1	+3	+3	+3	**+25**
Date duration	**+3**	**+6**	**+11**	**+23**	**+38**
Date recurrence	**+50**	**+81**	**+74**	**+72**	**+63**
Day of week	**+62**	**+59**	**+54**	**+47**	**+52**
Interval date	+2	+1	**+13**	**+13**	**+17**
Time addition	**+4**	**+14**	**+21**	**+40**	**+59**
Time subtraction	**+9**	**+11**	**+28**	**+48**	**+59**
Time duration	**+5**	**+25**	**+43**	**+41**	**+62**

Hindi: CoT minus direct

Short	Med.	Long	V. long	VV. long
+2	−2	+1	−6	**+21**
−1	**−9**	**−14**	**−31**	**−31**
−4	−2	+1	**+16**	**+41**
+19	**+38**	**+23**	**+6**	**+3**
+8	+5	+4	+5	**+10**
−36	**−28**	**−20**	**−17**	**−8**
−16	**−13**	**−14**	**+14**	**+32**
−9	**−7**	−0	**+20**	**+37**
−2	**+29**	**+44**	**+44**	**+49**

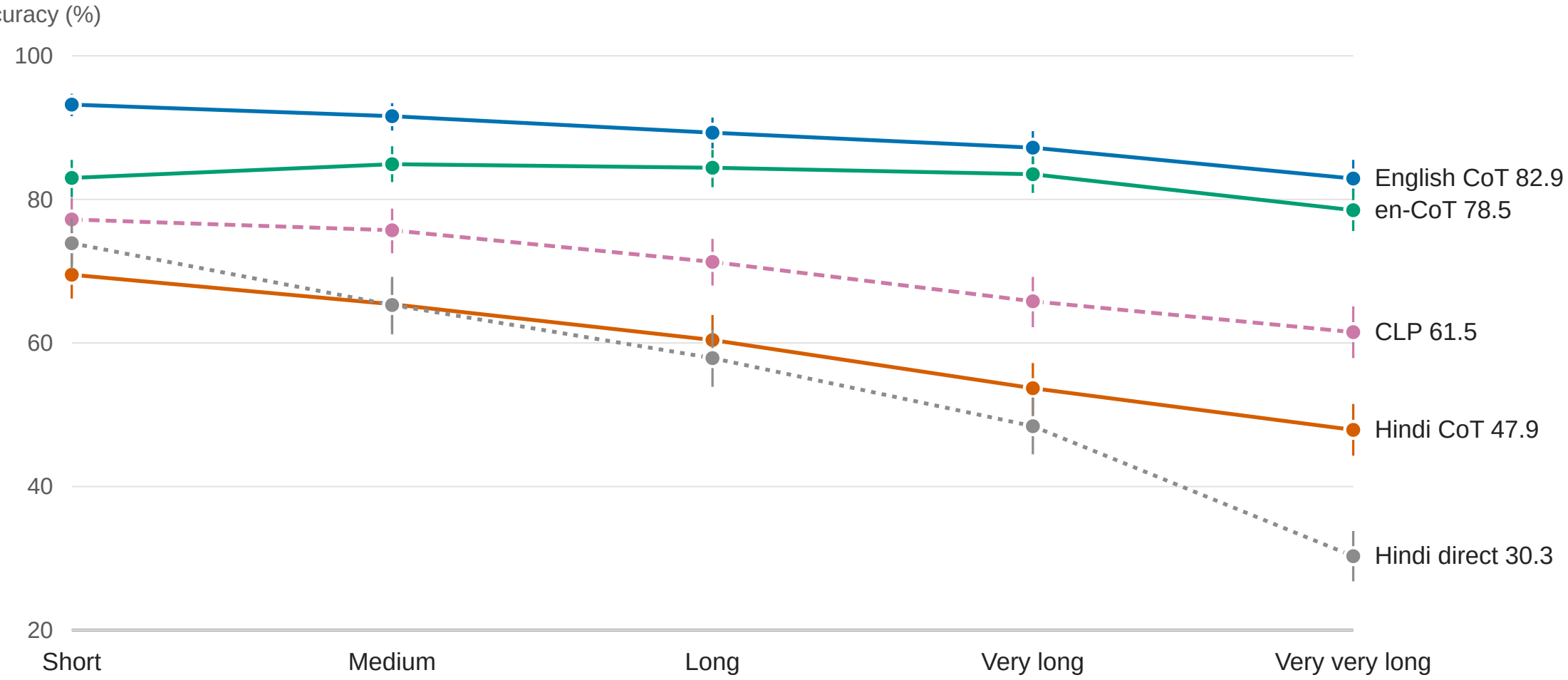
Bold: 95% interval excludes zero, uncorrected. 6,750 questions, answered, pooled.

RQ1: Two Hindi tasks get worse with CoT



2,250 questions, answered, pooled. All 6,750, no distractor: −19.9 [−23.4, −16.4] and −23.3 [−27.7, −19.0].

RQ1: The gap widens with difficulty, in points only



2,250 questions, 95% intervals. All 6,750: gap 24.0 → 34.9 (strict 23.0 → 31.1), trend p = 0.21.

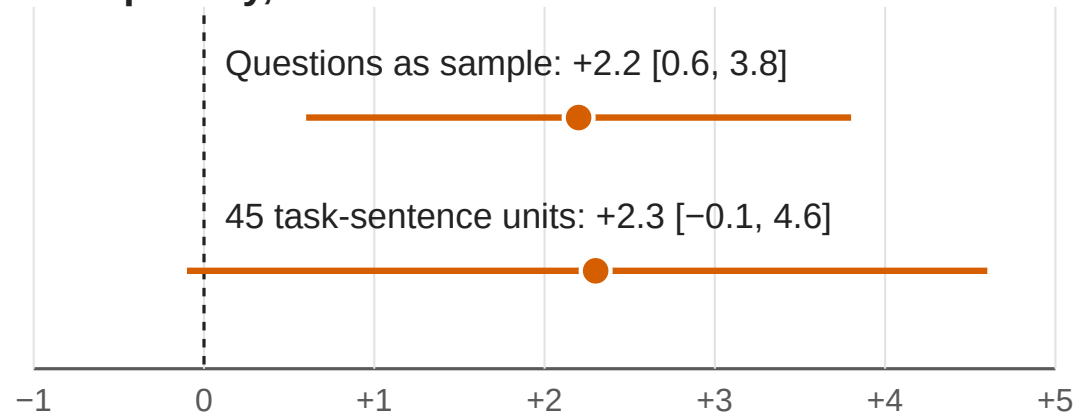
RQ2: The CoT penalty in Hindi is 2.2 points

Time-related distractor, drop in Hindi accuracy



CoT penalty: 2.2

CoT penalty, 95% interval



Leave one sentence out	penalty 1.8 to 2.6
Hindi minus English penalty	+2.6 [0.4, 4.7] answered, +1.4 [−0.7, 3.5] strict
English CoT drop	0.9 answered, 2.2 [1.3, 3.2] strict
Mid-report claim	withdrawn (fragility only in Hindi)

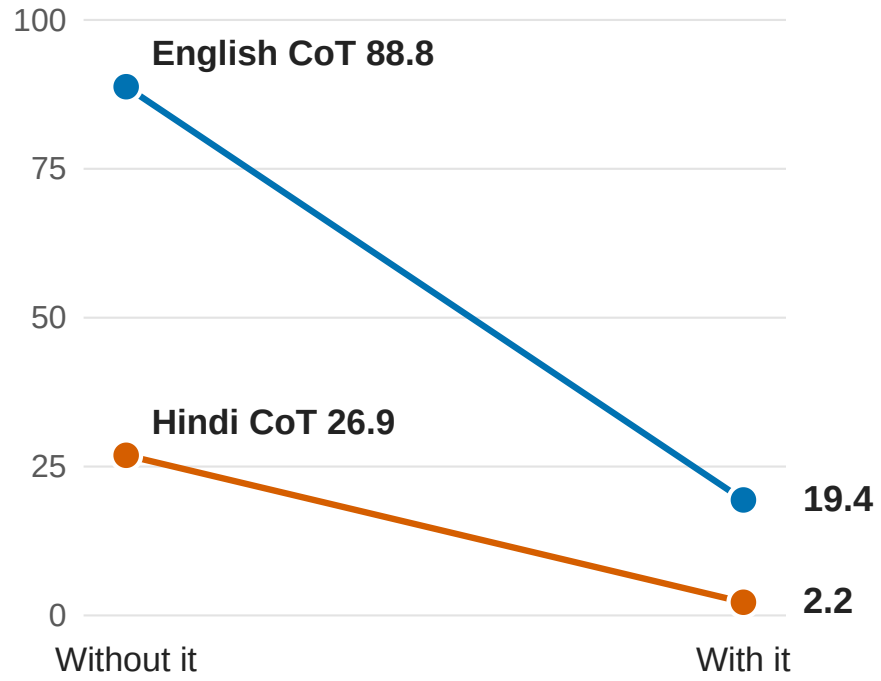
RQ2: Penalty by task

CoT penalty, time-related	English	Hindi	Hindi minus English
All	−0.3 [−1.6, 1.0]	+2.2 [0.6, 3.9]	+2.6 [0.5, 4.7]
All, strict	+0.8 [−0.5, 2.2]	+2.2 [0.6, 3.9]	+1.4 [−0.7, 3.6]
All, unrelated	−1.9 [−3.2, −0.7]	+0.7 [−1.0, 2.3]	+2.6 [0.4, 4.7]
Date addition	−0.1 [−2.6, 2.4]	−1.0 [−5.1, 3.2]	−0.9 [−5.7, 3.9]
Date subtraction	−1.7 [−4.9, 1.5]	−4.5 [−9.0, 0.1]	−2.8 [−8.0, 2.7]
Date duration	+2.3 [−0.6, 5.4]	+8.8 [4.2, 13.3]	+6.5 [1.4, 11.3]
Date recurrence	+8.4 [3.9, 12.7]	+4.5 [0.2, 8.6]	−4.0 [−10.0, 2.2]
Day of week	−1.5 [−7.4, 4.5]	+2.4 [−2.5, 7.2]	+3.8 [−3.7, 11.5]
Interval date	+3.2 [−2.7, 8.6]	−0.1 [−5.3, 5.2]	−3.2 [−10.4, 4.6]
Time addition	−6.1 [−9.9, −2.2]	+0.4 [−4.9, 5.7]	+6.5 [0.1, 13.1]
Time subtraction	−0.8 [−4.0, 2.6]	−1.9 [−7.1, 3.0]	−1.1 [−7.1, 5.1]
Time duration	−8.1 [−12.0, −4.2]	+11.5 [6.0, 17.5]	+19.7 [12.6, 26.9]

RQ2: Most of the harm comes from one sentence

> **"The last time I had a weekly commitment was in January."**

Strict accuracy, 134 date recurrence questions



English CoT cut-offs

4 → 92 of 134

Hindi CoT cut-offs **0**

Largest of 90 task-sentence units, found post hoc

None of the 92 cut-offs is a verbatim loop

Hindi recurrence penalty without it: +4.5 → +0.3 [−4.6, 4.9]

English recurrence penalty without it: stays +8.5

January competes with the question's start date

RQ2: The January sentence has the largest penalty in both languages

Time-related sentence	Task	Lang.	n	Cut-offs	Penalty
The last time I had a weekly commitment was in January.	Date recurrence	English	40	91	+23.7 [7.3, 41.1]
The last time I had a weekly commitment was in January.	Date recurrence	Hindi	134	0	+23.1 [15.7, 31.2]
Dates in February always confuse me a bit.	Date recurrence	English	155	12	+20.1 [11.3, 28.5]
Calendar dates can be so confusing sometimes.	Day of week	Hindi	120	7	+17.1 [6.2, 28.2]
I never know if my watch is running fast.	Time duration	Hindi	180	0	+16.1 [3.6, 28.1]
Time seems to move differently in spring compared to winter.	Date duration	Hindi	152	0	+15.8 [5.6, 25.9]
I never understood why we don't use decimal time.	Time duration	Hindi	121	1	+14.9 [2.3, 28.0]
Monday mornings are tough no matter what date it is.	Interval date	English	155	2	+13.5 [2.5, 24.6]
I wonder if counting backwards with dates is harder than forwards.	Date duration	Hindi	139	1	+11.4 [0.8, 22.4]

RQ2: Hindi traces mention distractors more often

Word list	Language	Distractor	Mention	Placebo	OR correct	95% interval
Stricter word list	English	Time-related	8.5%	0.2%	0.57	[0.39, 0.85]
Stricter word list	English	Unrelated	5.7%	0.1%	1.30	[0.62, 2.70]
Stricter word list	Hindi	Time-related	25.0%	0.2%	0.77	[0.67, 0.89]
Stricter word list	Hindi	Unrelated	29.6%	0.0%	1.15	[1.00, 1.33]
Original word list	English	Time-related	17.4%	9.9%	0.92	[0.73, 1.19]
Original word list	English	Unrelated	11.5%	5.9%	1.12	[0.82, 1.56]
Original word list	Hindi	Time-related	33.8%	10.1%	0.84	[0.73, 0.96]
Original word list	Hindi	Unrelated	30.4%	1.9%	1.13	[0.98, 1.30]

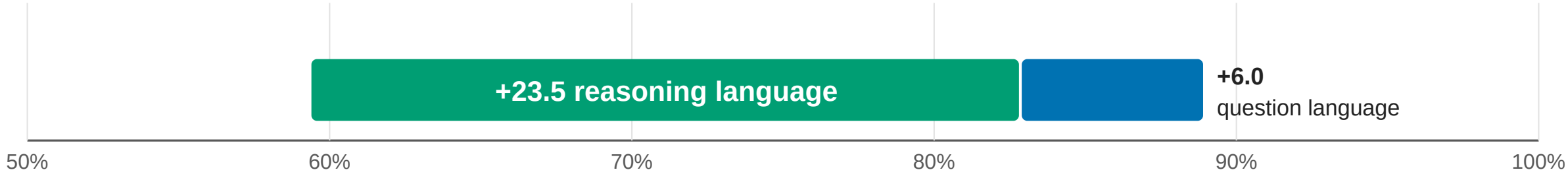
Literal reading After "My watch runs slow": "we cannot get 23:28 because my watch is slow". Answer 00:00.

RQ3: Most of the gap follows the reasoning language

59.4%
Hindi CoT
Hindi question
Hindi reasoning

82.9%
en-CoT
Hindi question
English reasoning

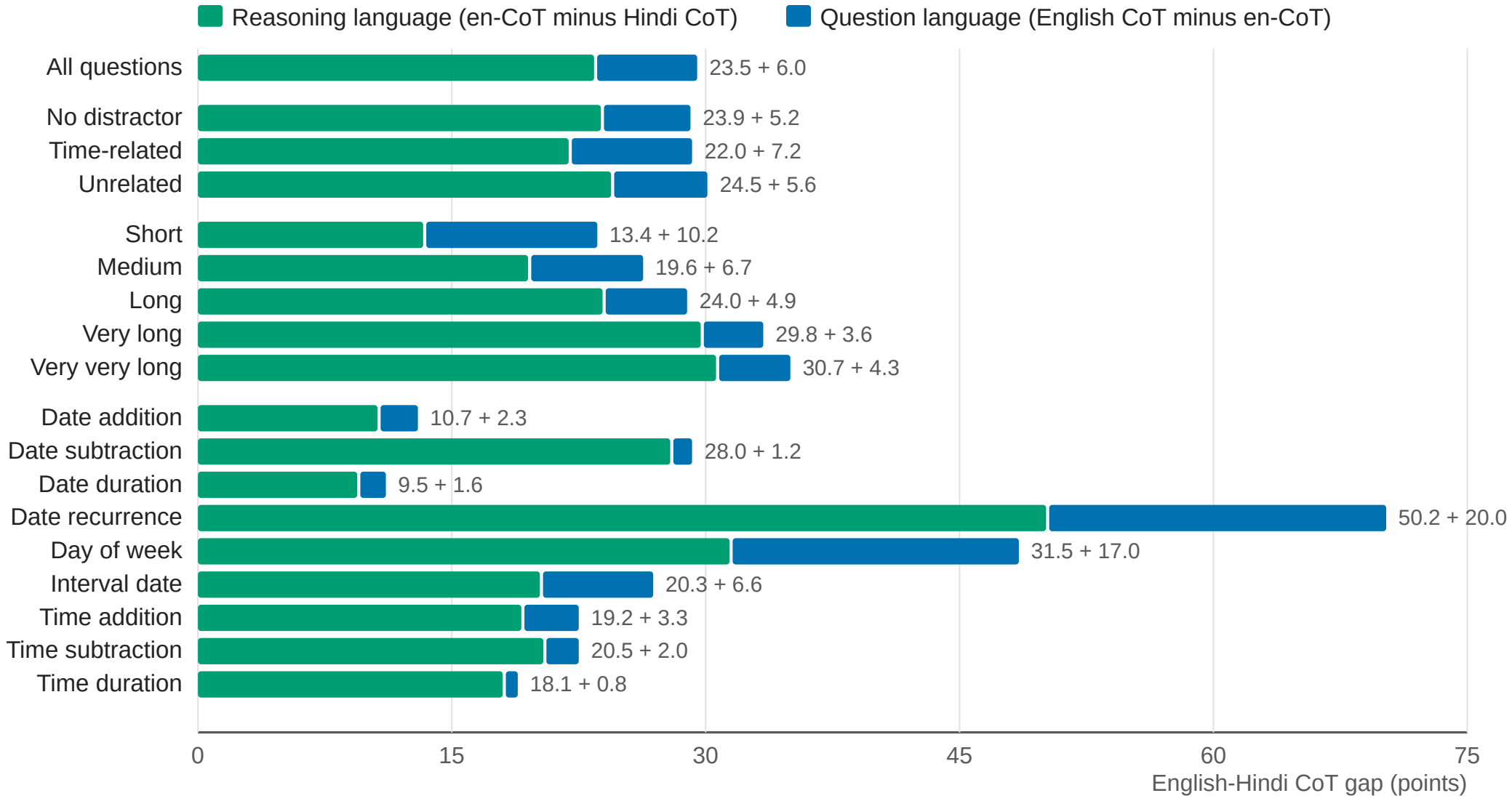
88.9%
English CoT
English question
English reasoning



	Points	95% interval
Total gap	29.5	[28.1, 30.9]
Reasoning part	23.5	[22.1, 24.9]
Question part	6.0	[5.0, 7.1]
Reasoning share	79.7%	[76.4, 82.9]

Strict reasoning share 80.7%
Direct gap, same questions 4.5
English Q + Hindi reasoning not run

RQ3: The reasoning part is larger on every task and distractor



RQ3: RDS and verification follow the reasoning language

Trace pair

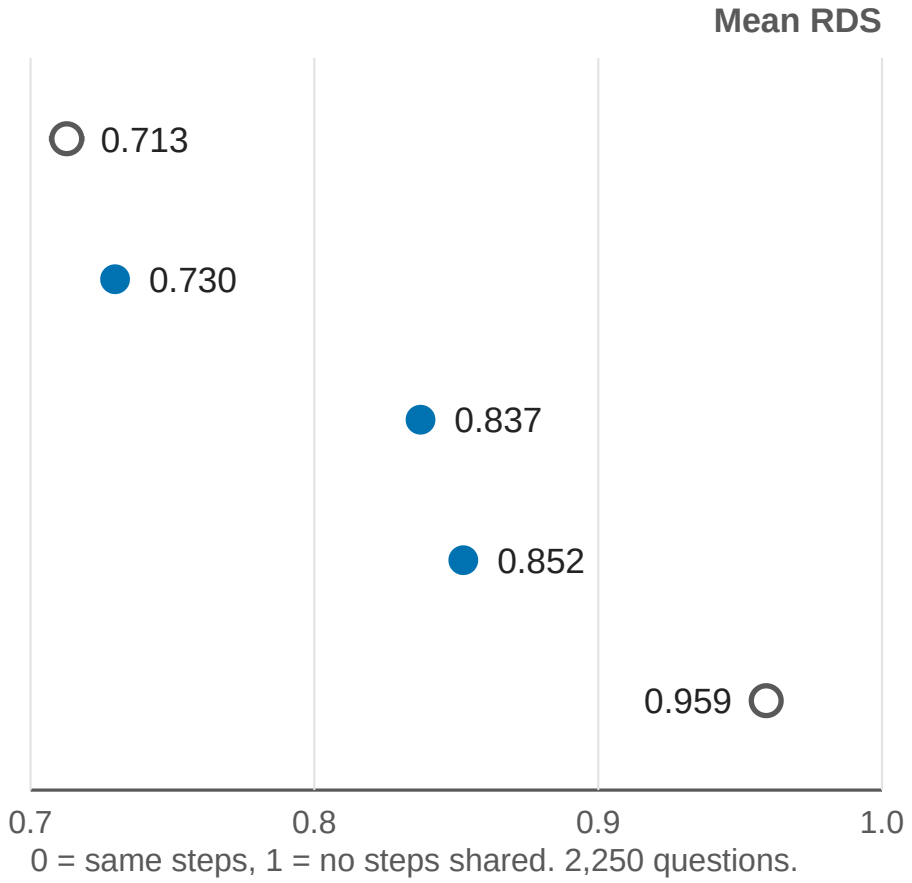
Distractor added
English CoT, with vs without

Question language
English CoT vs en-CoT

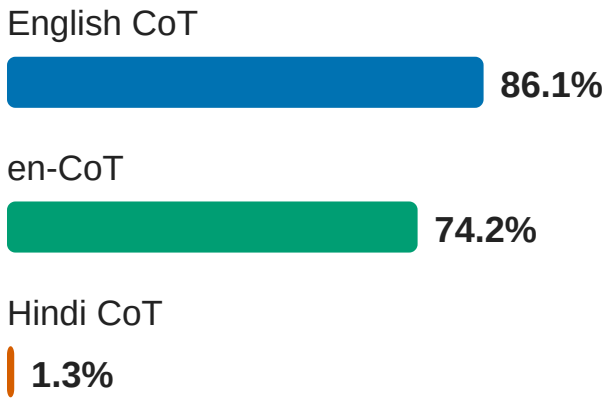
Reasoning language
en-CoT vs Hindi CoT

Question and reasoning language
English CoT vs Hindi CoT

Different questions
English CoT vs Hindi CoT



Traces with a verification phrase



Share of en-CoT gain from verification

1.0% [−6.3, 9.0], observational

RQ3: Hindi traces are shorter and rarely check their work

CoT traces	English	Hindi
Median steps	25	13
CHECK or REVISE step	88%	3.7%
Verification phrase, answered traces	86.1%	1.3%
Median tokens, answered	353	212

Hindi CoT language

0.988 mean TLR

2.14% of traces below 0.9

OR correct for those: 2.08 [1.66, 2.60]

Cut-off English CoT

42 check or revise steps on average, 8 on wrong traces that finish

29 February in span, longest questions, English CoT

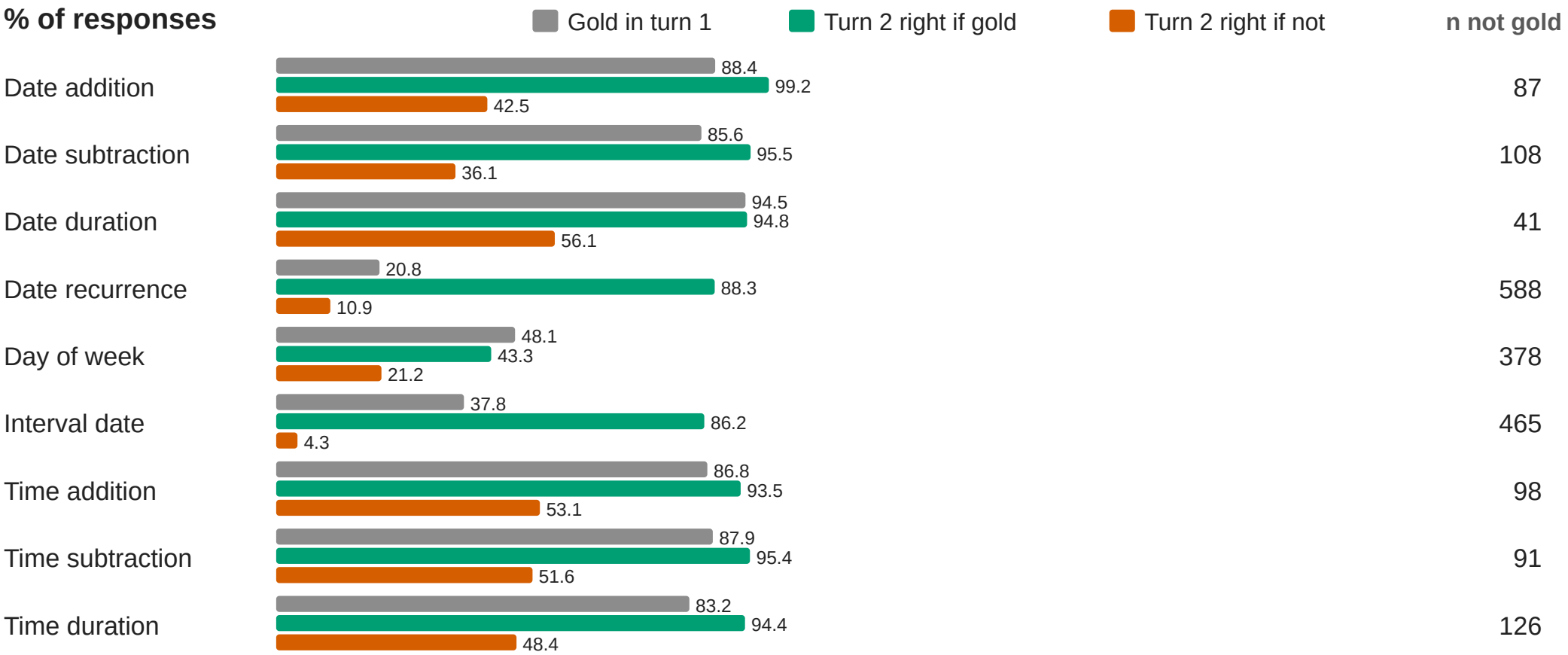
36% wrong, 9% without it. 181 of 186 wrong answers mention the leap year.

RQ3: CLP scores 70.3%, en-CoT 82.9%

2,250 Hindi questions	Direct	Hindi CoT	CLP	en-CoT	CLP − CoT	CLP − en-CoT
Date addition	82.8	85.1	92.7	95.7	+7.6	−3.1
Date subtraction	84.8	66.0	86.9	94.0	+20.9	−7.1
Date duration	78.9	88.2	92.7	97.7	+4.4	−5.1
Date recurrence	2.8	20.1	26.9	70.3	+6.8	−43.4
Day of week	15.2	21.0	31.8	52.5	+10.8	−20.7
Interval date	46.8	23.9	35.3	44.1	+11.4	−8.8
Time addition	75.6	73.5	88.2	92.6	+14.7	−4.5
Time subtraction	64.9	75.2	90.1	95.7	+14.9	−5.6
Time duration	44.7	80.1	86.7	98.1	+6.6	−11.5
No distractor	56.0	60.2	69.3	84.1	+9.0	−14.9
Time-related	54.9	58.9	69.3	81.0	+10.4	−11.6
Unrelated	54.6	59.1	72.3	83.6	+13.2	−11.2
All, answered	55.2	59.4	70.3	82.9	+10.9	−12.6
All, strict	55.2	59.0	69.9	80.8	+10.9	−10.9

CLP − Hindi direct: +15.1 [13.5, 16.7]. en-CoT − CLP: +12.6 [11.4, 13.9]. Qin et al. on MGSM: CLP 70.6, En-CoT 57.8.

RQ3: CLP turn 2 mostly follows turn 1



Three date-answer tasks: turn 2 right 97.5% if turn 1 ends at gold, 20.1% if not. Turn 1 not randomised.

ID 12. Turn 1: "the date of the 7th occurrence is 2 August 2027." Turn 2: "Adding 78 days to 16 May 2027 also results in 2 August 2027." Gold 2027-08-15.

RQ3: With thinking on, the gap falls to 1.0 point

	English CoT		Hindi CoT					
450 questions	Off	On	Off	On	Off strict	On strict	Cut 4,096	Over 4,096
Date addition	99.3	99.3	83.3	100.0	83.3	100.0	71.3	28.7
Date subtraction	96.7	98.7	68.7	100.0	68.7	100.0	66.0	34.0
Date duration	99.3	97.3	91.3	96.6	90.7	95.3	35.3	64.7
Date recurrence	87.6	98.3	16.0	93.4	16.0	75.3	17.3	82.0
Day of week	68.2	99.2	18.9	98.2	18.0	74.0	2.7	97.3
Interval date	56.0	97.6	27.7	94.8	27.3	73.3	0.7	98.7
Time addition	96.6	99.3	78.0	98.0	78.0	96.0	71.3	27.3
Time subtraction	98.7	99.3	73.8	98.6	73.3	97.3	63.3	35.3
Time duration	99.3	100.0	83.3	100.0	83.3	98.0	66.0	34.0

	Thinking off → on	
All	CoT, English	91.4 → 98.9
All	CoT, Hindi	64.1 → 97.9 (strict 59.9 → 89.9)
All	Direct, English	64.2 → 97.9
All	Direct, Hindi	59.2 → 94.9
Gap	CoT	27.3 → 1.0 (strict 27.4 → 2.7)

RQ3: Thinking on means 79% English reasoning and four times the tokens

79% English words in hidden Hindi CoT reasoning
Visible answer stays in Hindi

43.8% strict Hindi CoT cut at 4,096 tokens
59.9% with thinking off

Format arguments

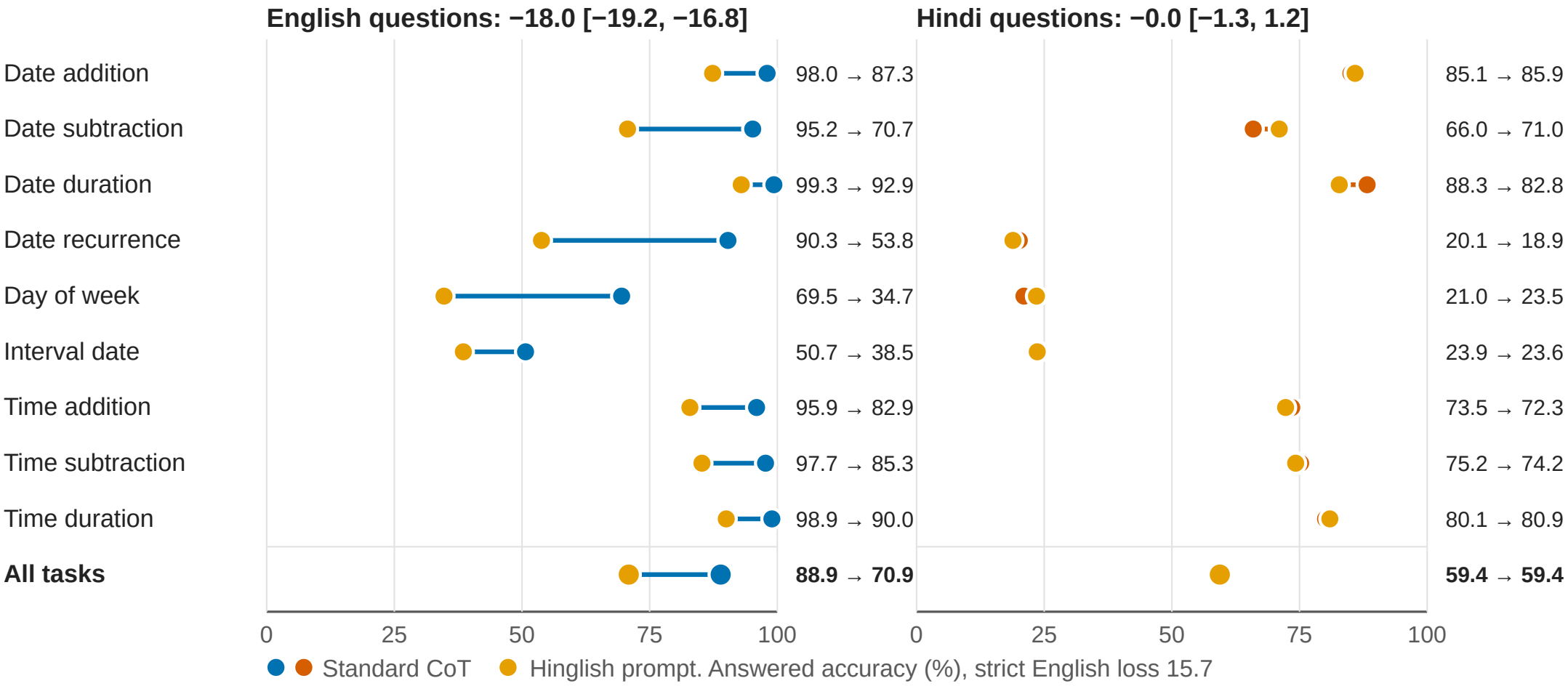
Hindi direct hidden steps that are FORMAT	207,097 of 627,487
Visible CoT steps that are FORMAT	0.2%
FORMAT share, date and time addition	49.8%
FORMAT share, interval date	15.3%
Mean FORMAT steps, right vs wrong traces	157 vs 120

Adding one day: 5,170 words, 361 of 636 FORMAT

"If I write: 2025-04-07. That's not in Hindi. ...
But if I have no text outside the last line,
then there's no Hindi text."

Errors	287 of 422 wrong responses have no answer line, mostly at the 16,000-token limit
Disagreement	31.7% → 10.9% on matched questions, unanswered counted wrong, OR 0.23 [0.18, 0.30]
Pairs	1,159 both right, 20 English only, 11 Hindi only, 1 both wrong

RQ4: Hinglish changes nothing on Hindi questions, −18.0 on English



RQ4: English loses most when subtraction crosses a month

Date subtraction	Crosses month	Same month
English CoT	90.1	100.0
Hindi CoT	36.2	93.8
en-CoT	88.1	99.5
Hinglish, English Q.	42.3	97.4
Hinglish, Hindi Q.	42.5	97.7
English Q., plain	65.1	100.0
English Q., Roman mix	35.0	96.5

No Hindi function words, English Q. n = 1,838

−10.3 [−12.3, −8.3] accuracy; verification 86.5% → 18.6%

Roman traces, English Q.: −11.3 [−13.6, −9.0] more

Roman vs Devanagari, Hindi Q.: −2.8 [−5.2, −0.5]

Script	Roman mix	Plain	Devanagari ≥10%
English Q.	63.1%	36.9%	0%
Hindi Q.	24.4%	2.6%	73.1%

Verification %	Regex	Extended
English CoT	86.1	86.5
en-CoT	74.2	74.7
Hindi CoT	1.3	1.3
Hinglish, English Q.	14.6	20.4
Hinglish, Hindi Q.	1.8	7.5

Majority Devanagari, Hindi Q.: 68.5%

Hindi Q. task swings: date duration −5.5, date subtraction +5.1

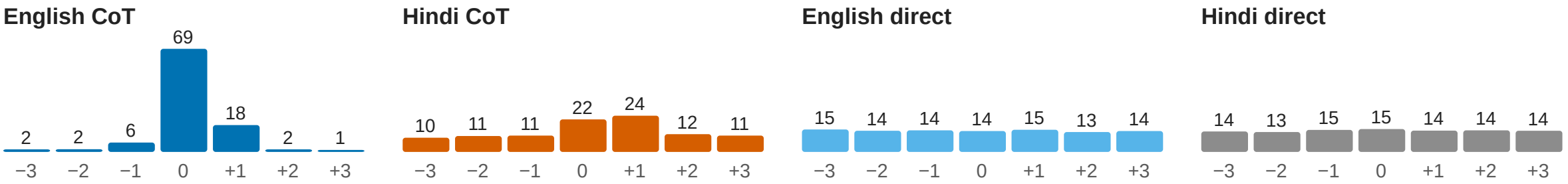
Errors: Three tasks hold most wrong answers

64% of wrong answers are day of week, interval date or recurrence (19,583 of 30,421)

78 to 88% when the model reasons in English

2,250 questions	Wrong	Day of week %	Interval date %	Verify %	Median tokens
English CoT	726	26.4	48.9	86.1	353
Hindi CoT	2,723	21.0	20.9	1.3	212
CLP	1,993	24.9	24.3	26.9	487
en-CoT	1,125	27.8	36.4	74.2	311

Day of week: answer minus gold, share of responses (%)

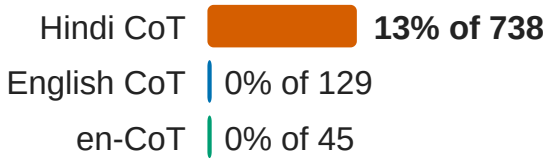


Errors: Some patterns come mainly with Hindi reasoning

Mainly when reasoning in Hindi

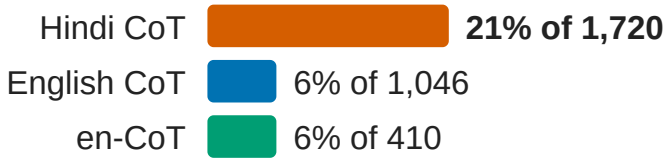
Date subtraction

answer on or after today



Interval date

span not seven days long



In either language

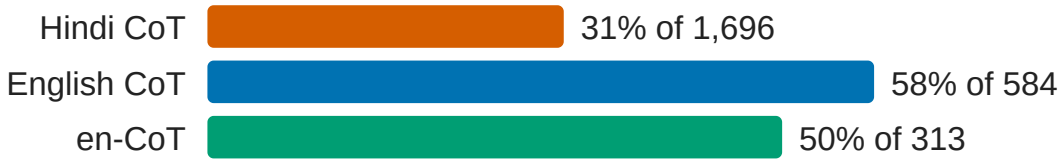
Date recurrence

one interval short



Day of week

exactly one day late



Shares of wrong answers of the expected form. Script labels, not hand-checked.

Errors: Hindi CoT examples

Hindi CoT, date subtraction: counts forward

Question 2027-05-09, 52 days ago

Trace From 9 May to 31 May is 22 days. ... April has 30 days, so the date exactly 30 days before 30 April is 1 May.

Answer 2027-05-01, gold 2027-03-18

Hindi CoT, interval date: two-day week

Question 2030-05-15, next week

Trace 15 May 2030 is a Saturday. ... The next Monday will be 18 May 2030. ... Sunday will be 19 May 2030.

Answer 2030-05-18 to 05-19, gold 05-20 to 05-26

Hindi CoT, time addition: 10:46 plus 19 hours 41 minutes

24 - 19 = 5 घंटे। तो, 19 घंटे और 41 मिनट का मतलब है कि हमें 5 घंटे और 41 मिनट जोड़ने हैं।

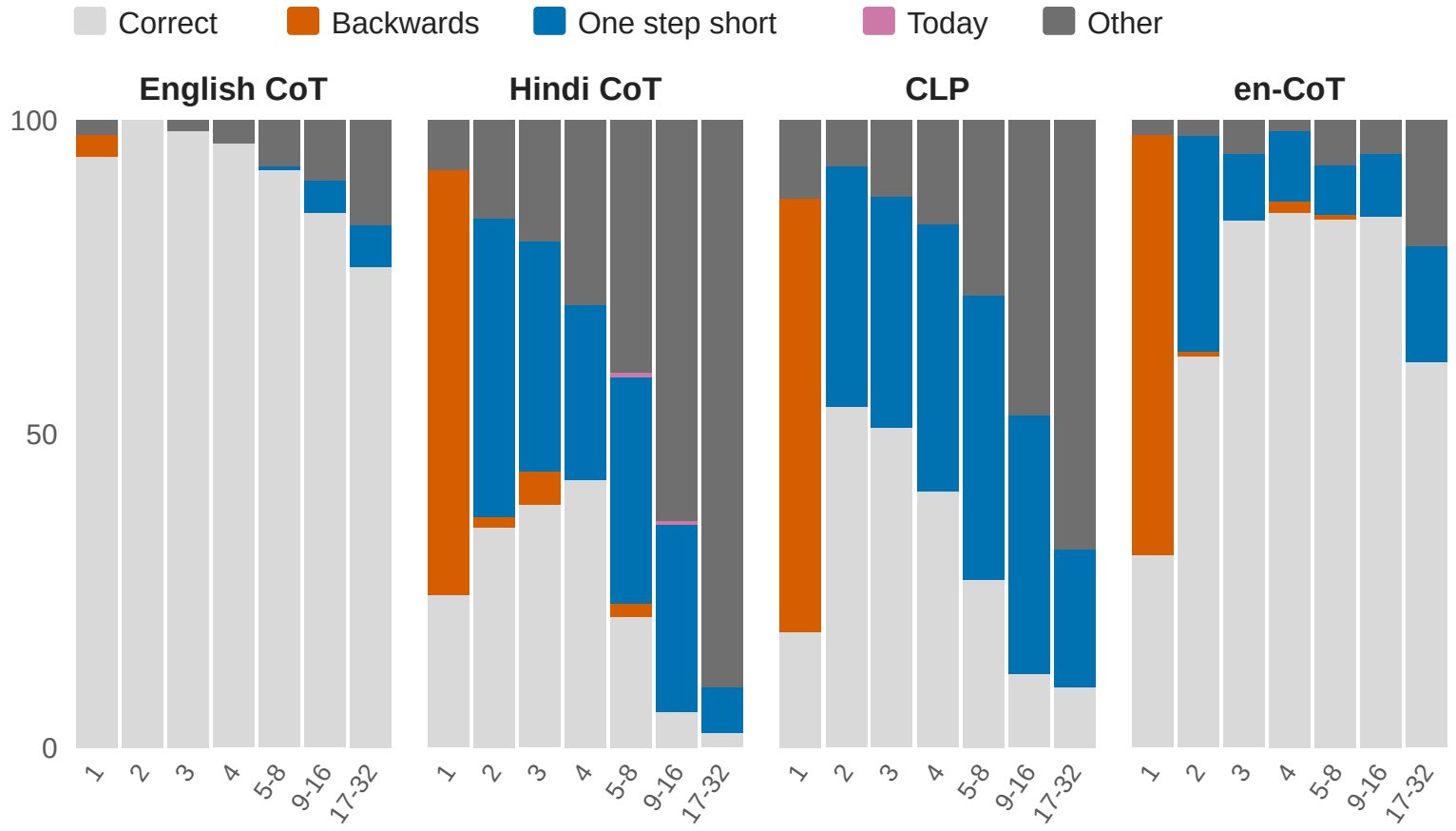
24 - 19 = 5 ghaṇṭe. to, 19 ghaṇṭe aur 41 minaṭ kā matlab hai ki hameṃ 5 ghaṇṭe aur 41 minaṭ joṛne haiṃ.

'24 - 19 = 5 hours. So 19 hours and 41 minutes means we have to add 5 hours and 41 minutes.'

Answer 16:27, gold 06:27.

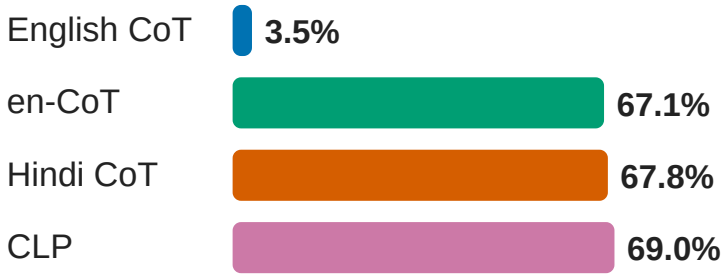
Time addition error rate: 24 to 27% at every level.

Errors: en-CoT keeps the errors from reading Hindi



Occurrence number k. Share of answers (%), 2,250 questions.

Backward answers at k = 1



आज को छोड़कर, 1वीं आवृत्ति
"excluding today, the 1st occurrence"
en-CoT: "this suggests we need to find what comes before"

1वीं is non-standard. No corrected-template rerun. k = 2 to 4: 2 of 229 en-CoT backward. "0 hours" wrong: en-CoT 19%, others 5%.

Errors: Other examples

English CoT, day of week: one day late

Question Day of week of 2029-07-02, gold Monday

Trace Total: 31 + 28 + 31 + 30 + 31 + 30 + 2 = 183 days.

Answer Tuesday. 1 January to 2 July is 182 days.

English CoT, recurrence: skips 29 February

Question 2028-02-15, every 6 days, 4th occurrence

Trace From 2028-02-27 to 2028-03-05: 28 - 27 = 1 day in February, plus 5 days in March = 6 days ✓

Answer 2028-03-11, gold 2028-03-10

en-CoT, recurrence at k = 1: goes backwards

Question आज को छोड़कर, 1वीं आवृत्ति, ID 5

Trace Given the phrasing "excluding today," this suggests we need to find what comes before.

Answer 2025-04-06, gold 2025-04-14

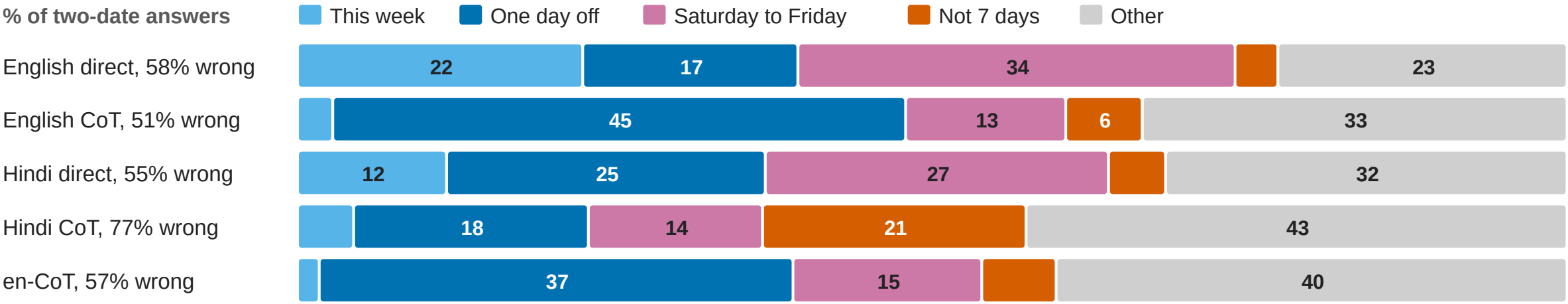
Hinglish, English question: date subtraction

Question 2025-05-01, 3 days ago

Trace 01 minus 3 karne par hum 31 April par aayenge.

Answer 2025-04-30, gold 2025-04-28

Errors: Next-week answers go wrong differently by condition



Skips a month end (en-CoT, today 2027-05-27)

Trace This week ends on Sunday, 2027-05-30. ... Next week starts on Monday, 2027-06-01.

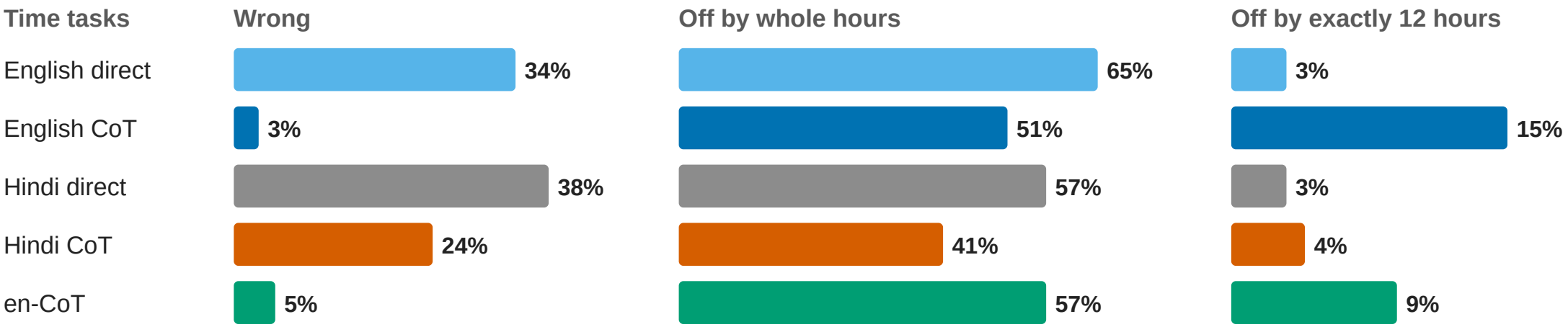
Answer 2027-06-01 to 06-07, gold 05-31 to 06-06

Gives this week (English direct, today 2026-09-13)

Why 13 September 2026 is a Sunday, so the answer is the week ending today.

Answer 2026-09-07 to 09-13, gold 09-14 to 09-20

Errors: Time answers get the minutes right and the hours wrong



12-hour clock (English CoT): 07:22, 15 h 31 min ago

Trace 6 - 15 = -9 hours. Since I have -9 hours, I need to go back 12 hours to get into the previous day: -9 + 12 = 3 hours.

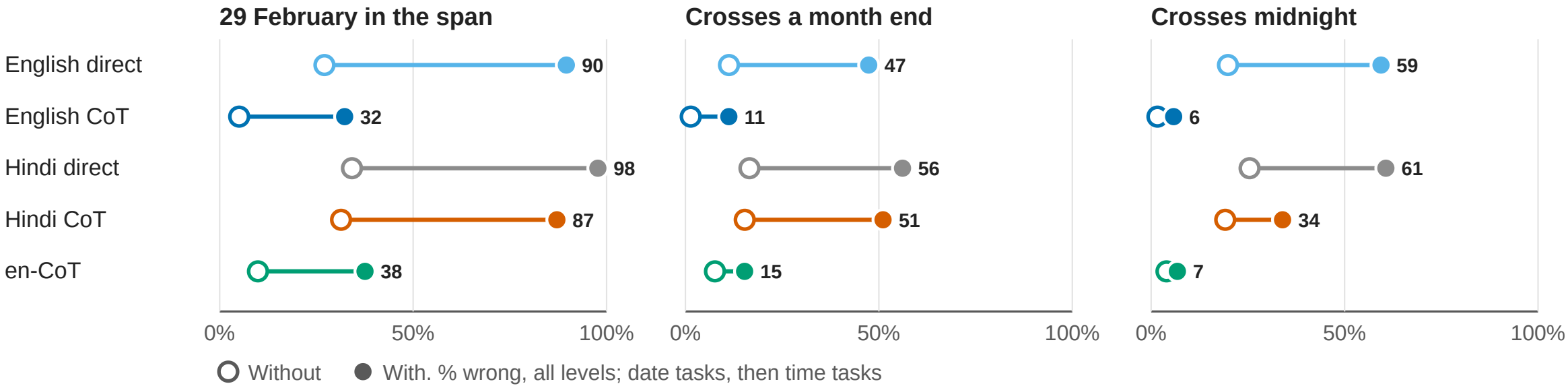
Answer 03:51, gold 15:51

"0 hours" misread (en-CoT): 12:11, 0 h 4 min ago

Trace I need to find what time it was 4 hours and 4 minutes before 12:11.

Answer 08:07, gold 12:07

Errors: Crossing a boundary raises the error rate



Knows the leap year, skips the day anyway (Hindi CoT)

Trace 6 दिन बाद: 6 मार्च 2028 ... चूंकि 2028 एक लीप वर्ष है

Answer 2028-03-06, gold 2028-03-05

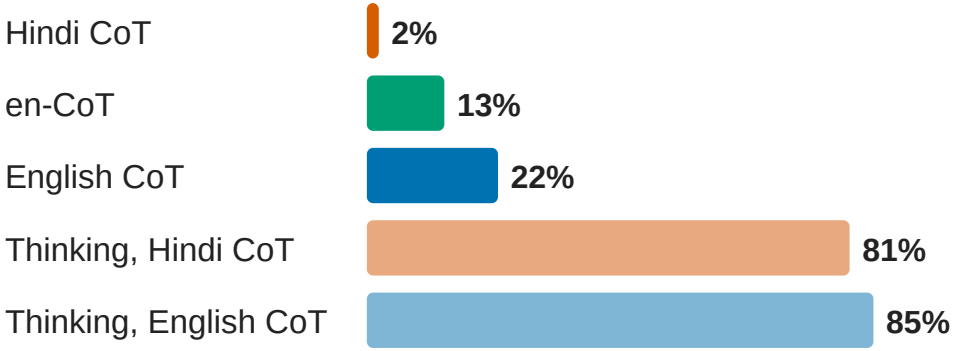
Check repeats the mistake (English CoT)

Trace From 2028-02-27 to 2028-03-05: 28 - 27 = 1 day in February

Answer 2028-03-11, gold 2028-03-10

Errors: English CoT's cut-offs average 42 re-checking steps

Wrong answers that are cut-offs



English CoT's 651 cut-offs



Day of week, recurrence, next week, other

CHECK + REVISE steps, English CoT	Wrong	Right
Day of week	23.0	13.4
Date recurrence	19.3	5.0
Time addition	13.1	3.5
Date duration	12.6	2.8
Time duration	12.4	2.0
Time subtraction	11.7	3.3
Interval date	11.2	7.7
Date subtraction	6.0	2.5
Date addition	2.4	1.1

Wrong traces with 20 or more: 869 of 2,903

English CoT, day of week of 2025-08-21, gold Thursday, no answer

Day 233: (233-1) mod 7 = 1. Wednesday + 1 = Thursday. But month-by-month gave Friday.
There's a 1-day discrepancy. Let me recheck July 31 ...

Errors: Distractors can steer the answer or the reasoning

Answer lands on the month end (English direct)

Question I always get confused with months that have 31 days. Today is 2026-12-24, what is the date going to be in 8 days?

Answer 2026-12-31, gold 2027-01-01

Pulled into the named month (Hindi direct)

Question अगस्त मेरा पसंदीदा महीना है। आज 2025-07-29 है, 1 दिनों में तिथि क्या होगी?

English August is my favourite month. Today is 2025-07-29. What is the date in 1 day?

Answer 2025-08-01, gold 2025-07-30

Changes a fact (Hindi CoT, 2026-09-02, 12 days ago)

Trace 31 दिनों वाले महीने अप्रैल, जून, सितंबर और नवंबर होते हैं।

English The months with 31 days are April, June, September and November.

Answer 2026-08-31, gold 2026-08-21

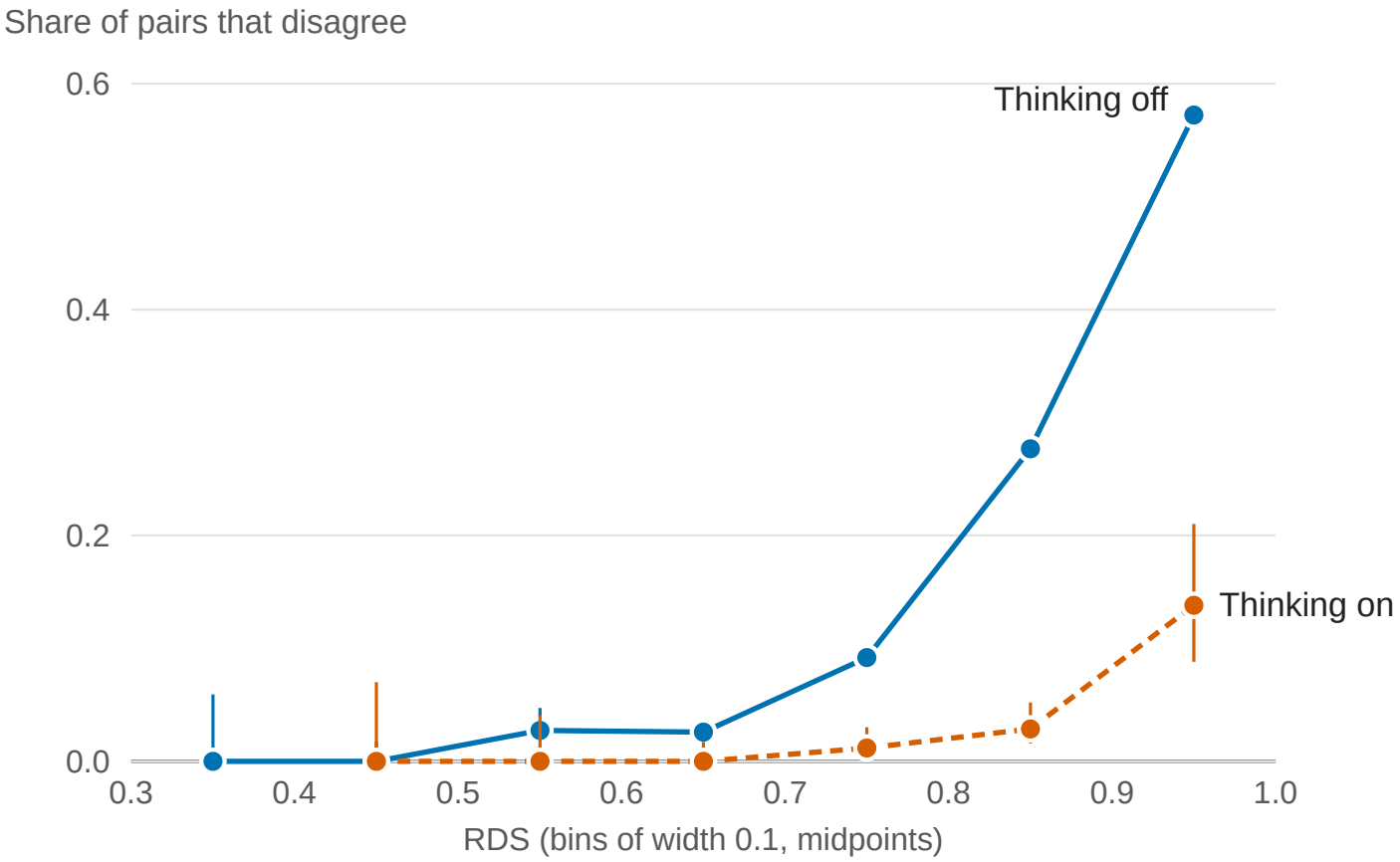
Argues until the token limit (English CoT)

Question My watch always runs a bit slow. It is now 13:44, what was the time 59 hours and 37 minutes ago?

Trace Could 'it is now 13:44' refer to the watch?

Answer none, gold 02:07

RQ5: More divergent traces go with disagreement



31.4% of pairs disagree

0.917 vs 0.809 mean RDS, disagree vs agree

Odds ratio per +0.1 RDS, adjusted

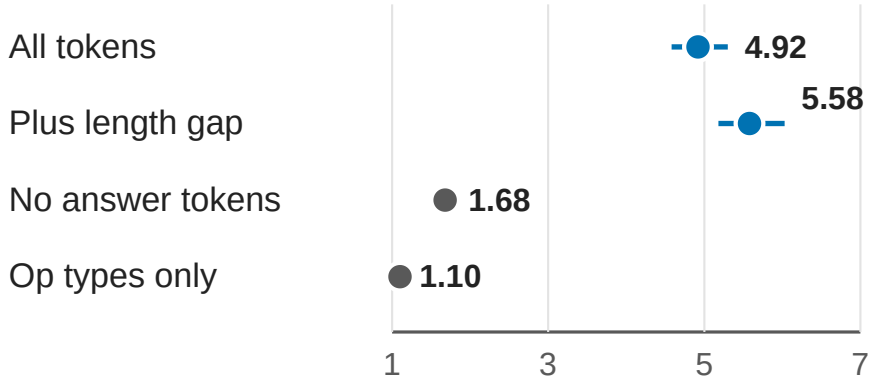


Chart is unadjusted; bins under 20 pairs hidden. In-sample association. 88% of pairs above RDS 0.7.

RQ5: Removing answer tokens takes the OR from 4.92 to 1.68

Token view	OR/0.1	95%	OR/SD	95%	AME	AUC
All tokens	4.92	[4.58, 5.30]	6.47	[5.94, 7.04]	22.9	0.853
No answer tokens	1.68	[1.60, 1.76]	1.65	[1.58, 1.73]	9.0	0.771
No answers, no answer values	1.51	[1.45, 1.58]	1.52	[1.45, 1.58]	7.2	0.767
Op types	1.14	[1.11, 1.18]	1.21	[1.16, 1.26]	2.4	0.756
Op types, no answers	1.10	[1.07, 1.13]	1.14	[1.09, 1.18]	1.7	0.755
First 8 tokens	1.16	[1.12, 1.19]	1.20	[1.15, 1.24]	2.6	0.758
First 8 op types	1.04	[1.02, 1.06]	1.07	[1.03, 1.10]	0.7	0.755
Before first check	1.32	[1.28, 1.37]	1.41	[1.36, 1.47]	4.9	0.766
Op types before first check	1.06	[1.03, 1.08]	1.08	[1.04, 1.12]	1.0	0.756

Compare final answers alone AUC 0.951, recall 1.00, precision 0.82

Step types added out of fold AUC +0.0007 [−0.0004, 0.0018]

RQ5: By outcome and task

Thinking off	Pairs	RDS	No answers	Op types	En len	Hi len	Len gap
Both right	11,309	0.798	0.862	0.712	18	11	0.413
English only	5,947	0.917	0.910	0.763	41	15	0.555
Hindi only	505	0.925	0.920	0.779	58	15	0.581
Both wrong	1,710	0.925	0.927	0.816	79	19	0.678

By task	Disagree	OR/0.1 all	No answers	Op types
Date addition	15.8%	6.99	1.45	0.97
Date subtraction	31.6%	8.44	1.54	0.97
Date duration	13.2%	16.06	2.18	1.10
Date recurrence	68.2%	4.02	2.32	1.48
Day of week	58.2%	1.83	1.43	1.22
Interval date	45.5%	2.40	1.49	0.98
Time addition	26.3%	4.46	1.46	1.11
Time subtraction	25.2%	5.32	1.38	1.02
Time duration	21.9%	10.40	2.41	1.18

RQ6: Deleting the mention helps no more than a neutral deletion

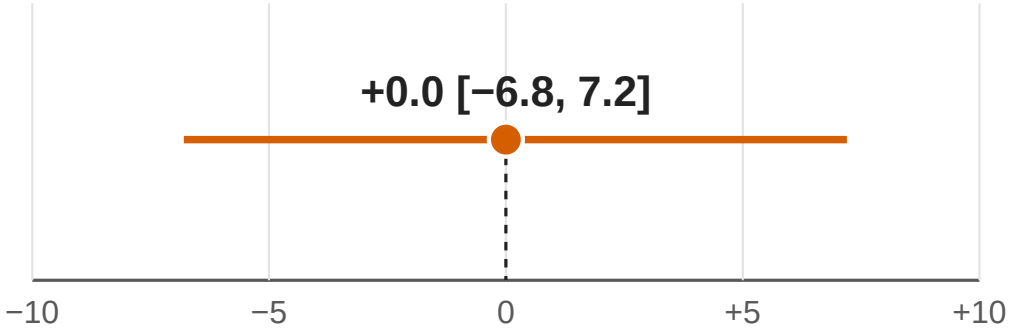
Three versions of each trace, one greedy continuation each

A	as written	s0	s1	s2	s3 (k)
B	mention deleted	s0	s1	s2	s3 (k)
C	neutral sentence deleted	s0	s1	s2	s3 (k)

234 traces, 222 Hindi.

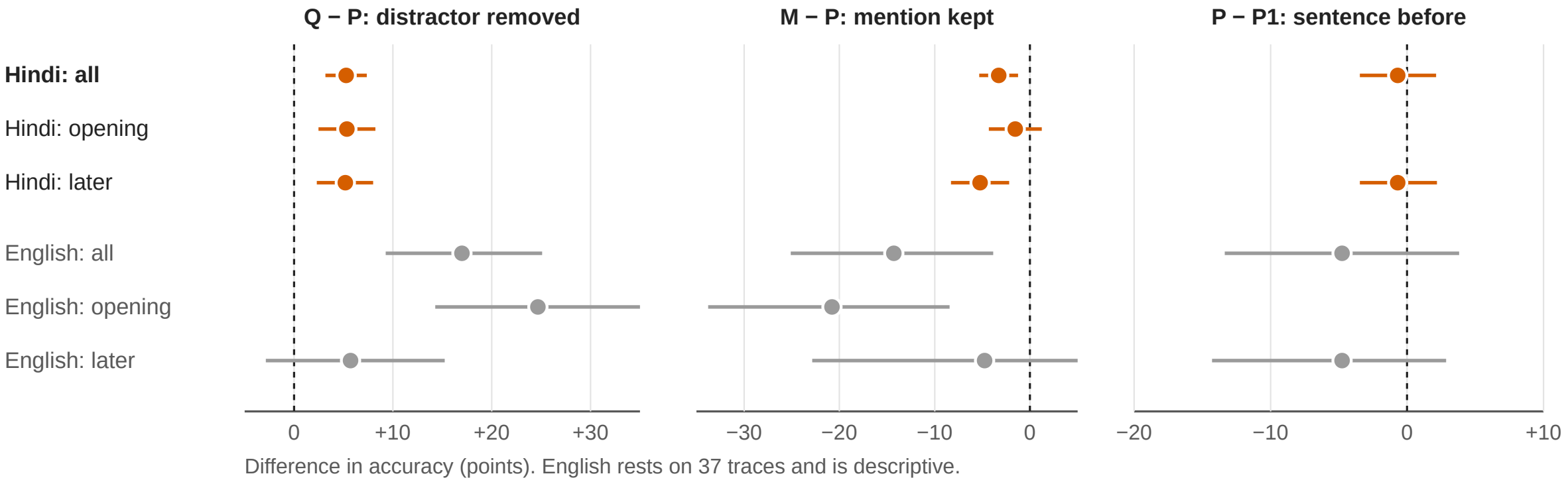
Deletion with sampling: +4.5 [−2.7, 11.7]

B minus C, points, 95% interval



Power for a 5-point effect: about 24%

RQ6: Removing the distractor helps a little



Difference in accuracy (points). English rests on 37 traces and is descriptive.

+5.3 [3.2, 7.4]

Q − P in Hindi: 45.2% vs 40.0%. With the 29 sentences as the sample: [−0.3, 9.4].

−3.3 [−5.4, −1.2]

M − P in Hindi. Not clearly more than an ordinary preceding sentence (p = 0.09).

58%

of the Q − P effect comes from the January sentence. Without it: +2.8 [0.4, 5.0].

RQ6: The mention is not distinguishable from an ordinary sentence

	Correct (%)				Unanswered (%)			
	Q	P	M	P1	Q	P	M	P1
Hindi, all	45.2	40.0	36.7	28.6	5.7	10.0	10.8	13.8
Hindi, opening	56.0	50.7	49.1		3.5	6.9	8.1	
Hindi, later	33.1	27.9	22.7	28.6	8.1	13.4	13.7	13.8

Hindi contrasts	
Excess fraction, Hindi	8.8% [5.5, 12.1]
Q − P, sentences as sample	+5.3 [−0.3, 9.4]
Q − P without January sentence	+2.8 [0.4, 5.0]
M − P	−3.3 [−5.4, −1.2]
P − P1	−0.7 [−3.5, 2.1]
(M − P) − (P − P1), 248 traces	−4.6 [−9.6, 0.6], p = 0.09
Rollouts mentioning the distractor	57.8%

Self-hosted minus API	
Hindi, 278 questions	+5.7 [0.5, 10.8]
English, 22 questions	−24.7
Calibration, 90 questions	−2.2 [−13.3, 7.8]

RQ6: One trace, every variant

Long date addition, ID 11, Hindi, translated

August is my favourite month. Today is 2026-07-24; what will the date be in 8 days? Gold 2026-08-01.

s0 First, we have to understand the given information carefully. s1 Today's date is 2026-07-24.

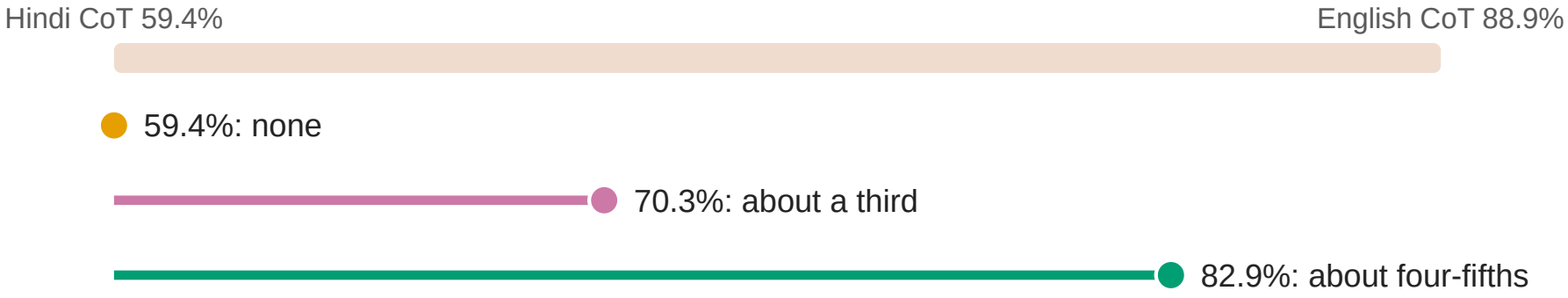
s2 The question asks what the date will be 8 days from today. s3 (k) The question also says that August is my favourite month.

Later: "One day past 31 July is 1 August, and the remaining 7 days give 8 August." Answer 2026-08-08.

Variant	Distractor	Prefix	Correct
A as written	with	s0 to s3	1 of 1
B mention deleted	with	s0 to s2	1 of 1
C neutral deleted	with	s1 to s3	1 of 1
P before the mention	with	s0 to s2	6 of 7
Q distractor removed	without	s0 to s2	7 of 7
M after the mention	with	s0 to s3	7 of 7
P1 a sentence earlier	with	s0 to s1	6 of 7

Main findings

29.5-point gap on Hindi questions: share closed by each prompt



CoT gain: +29.5 English, +4.7 Hindi; gap 4.3 → 29.1

Thinking on, 450 questions: gap 27.3 → 1.0, with four times the tokens

Hindi losses: methods seen mainly in Hindi traces; errors from reading Hindi remain under en-CoT

Distractors: penalty 2.2, largely one sentence

RDS: OR 4.92 per 0.1, association only; 1.10 on step types without answers

Interventions: Q − P +5.3; written mention not isolated

Limitations

Scope	One model, one prompt per condition, one seed, two languages.
Causal claims	Language and prompt wording confounded. No English question with Hindi reasoning.
Data	2,250-question subset for added sets. Hindi distractors not literal.
Measurement	Regex parser and script labels. One LLM annotator, no human validation.
Interventions	Wrong traces only. Quantised self-hosted model.
Thinking	10 questions per cell. Thinking and budget change together.

Future work

Limit	Next run
Non-standard Hindi carriers	Rerun recurrence with corrected templates
No English questions with Hindi reasoning	Add that arm
One prompt per condition	Several paraphrases of each prompt
CLP's first turn often solves the problem	CLP with solving forbidden in stage 1
Thinking and token budget changed together	Thinking on and off at equal budgets
Interventions cover wrong traces only	Question-by-mention factorial on correct and wrong traces
Labels read by script and one LLM annotator	Human annotation of samples and error labels

Thank you

Questions?

Team Netherite